MILITARY SPECIFICATION

JACK, HYDRAULIC, TRIPOD MMU-107/E; 40-TON CAPACITY

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 40-ton capacity, hydraulic tripod jack, designated MMU-107/E.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-104	Crate, Wood, Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-C-52950	Crate, Wood, Open and Covered
MIL-H-81019	Hydraulic Fluid, Petroleum Base, Ultra-Low Temperature

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-794	Part and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods & Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-1186 Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWING

Regent Jack Manufacturing Company, Incorporated.

8215 Jack, Hydraulic Tripod, Type MMU-107/E

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* 2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article testing. (See 4.4 and 6.3.)

3.2 General requirements. The requirements of Drawing 8215 apply as requirements of this specification.

3.3 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.4 Workmanship. The jack, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, sharp edges, accuracy of dimensions, and marking of parts and assemblies.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the prformance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. The inspection and testing of the jack shall be classified as follows:

a. First Article . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions.

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches Hg at a temperature of 25°±5°C and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the changes in instrument readings.

4.3.2 Horizontal load tests. Horizontal loads shall be applied toward an opposite leg. Twenty cycles shall be directed toward each leg. The jack shall then be rotated 120°. Horizontal loads shall be applied to the jack through the jack pad socket adapter.

4.3.3 Hydraulic fluid pumping source. The integral hydraulic pumping system shall be activated for all load lifting tests specified herein, except for the test specified in 4.6.5.3.

4.4 First Article testing (see 6.3).

4.4.1 Test sample. One jack shall be subjected to the first article tests specified in 4.4.3.

4.4.2 Test report. Upon completion of the first article tests, a test report shall be prepared in accordance with MIL-STD-831.

4.4.3 First Article tests. First Article tests shall consist of all tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of:

a. Individual tests.

b. Sampling plan and tests.

4.5.1 Individual tests. Each jack shall be subjected to the following tests as described under 4.6:

a. Examination of product.

b. Hydrostatic pressure and leakage.

c. Bypass valve test.

4.5.2 Sampling plan and tests.

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4.5.2.1 Periodic sampling. Five jacks shall be selected at random from every 50 or fraction thereof and subjected to the static proof test described in 4.6.4.

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.1.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a test failure. But final acceptance of the items on hand or items produced later shall not be made until is is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The jack shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, and compliance with the drawing.

4.6.2 Initial inspection. The jack shall be disassembled to the maximum practical extent and completely inspected. Each component shall be checked as to weight, dimensions, finishes, et cetera.

4.6.3 Operational test. The jack shall be operated under the following conditions to insure that the performance of the various components comply with the requirements specified herein. The jack shall be serviced with hydraulic fluid conforming to MIL-H-5606 unless otherwise specified.

4.6.3.1 Hydrostatic pressure and leakage. The ram and cylinder assembly shall be tested by applying an internal hydraulic pressure of 5,000 psi to the assembly. This test shall be performed with an appropriate load or restraint on the ram to preclude damage to the ram cylinder stops. There shall be no evidence of external leakage or deformation of assembly or component parts.

4.6.3.2 Settling. With the jack in the fully extended position and supporting rated capacity load (ram locknut placed in nonoperating position), the jack shall be allowed to stand for 1 hour. The height shall be measured at the beginning and at the end of this period. The rate of settling of the jack shall not exceed 0.020 inch per hour under rated capacity load, any excess of this distance shall be cause for rejection.

4.6.3.3 Bypass valve. Each pump shall be tested by sealing the pump outlet and operating the pump until the bypass valve opens. The bypass valve shall be adjusted to 105 to 110 percent of the fluid pressure required to raise a capacity load.

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4.6.3.4 Ram locknut. The ram locknut shall be tested with the jack ram at the fully extended position and supporting a rated capacity load. The hydraulic system shall be instantaneously vented to simulate a hydraulic failure. The locknut shall hold without dropping the load. Failure to hold the load shall be cause for rejection.

4.6.4 Static proof test. After the bypass relief valve has been adjusted for overload, the jack shall be tested with the screw extension in the fully extended position by raising a 60-ton load vertically from the lowest position of the hydraulic ram to the fully extended position. While in the extended position, the vertical load shall be reduced to 40 tons and maintained during application of a 6-ton horizontal load applied to the top of the extended screw extension. To complete the cycle, the horizontal and vertical loads shall be removed and the jack ram allowed to retract to its original position with not more than 50 pounds applied to the top of the ram. There shall be no evidence of external leakage of hydraulic fluid from the ram or cylinder assembly. Failure of any part of the jack during this test shall be cause for rejection.

4.6.5 Life test. With the screw extension retracted, the jack shall be subjected to the following tests:

4.6.5.1 Vertical lift. The vertical lift test shall consist of 200 cycles of raising and lowering a vertical load of 40 tons. One cycle shall consist of raising the hydraulic ram under load from the fully retracted position to the fully extended position and return to the fully retracted position. Failure of any part of the jack during this test shall be cause for rejection.

4.6.5.2 Dynamic side load. The dynamic side load test shall consist of 60 cycles of raising a 40-ton load vertically with a graduated 6-ton horizontal load. Twenty cycles shall be directed toward an opposite leg of the jack in accordance with 4.3.2 The horizontal load on the jack adapter shall be gradually increased from zero in the fully retracted position to 6 tons in the fully extended position. Failure of any part of the jack during this test shall be cause for rejection.

4.6.5.3 External pumping. Ten cycles of lifting and lowering a capacity load of 40 tons shall be conducted with hydraulic fluid pressure furnished through the external pumping source connector. Failure of any part of the jack during this test shall be cause for rejection.

4.6.6 Environmental tests. The following environmental tests shall be conducted in accordance with the specified methods of MIL-STD-810:

4.6.6.1 Low temperature. The jack shall be subjected to low temperature in accordance with method 502, procedure I. At the end of the exposure and with the temperature maintained, the jack ram shall be raised to its maximum extension and pumped until the bypass relief valve operates. The lowering valve shall be opened and the ram shall retract with not more than a 50-pound force applied to the ram. For this test, the jack may be serviced with hydraulic fluid conforming to MIL-H-81019. The number of full strokes necessary to fully extend the ram shall be recorded at both +70° and -65°F. The full strokes required to fully extend the ram at -65°F shall be not more than double the strokes required at +70°F.

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4.6.6.2 High temperature. The jack shall be subjected to high temperature in accordance with method 501, procedure I. At the end of the exposure and with the temperature maintained, the jack ram shall be raised to its maximum extension and pumped until the bypass relief valve operates. The lowering valve shall be opened and the ram shall retract with not more than a 50-pound force applied to the ram. During this test, there shall be no evidence of external leakage of hydraulic fluid from the ram-cylinder assembly.

4.6.7 Mobility. The jack shall be towed over improved roads for a distance of 1 mile to determine the general functioning of the running gear and towing capability. The test shall include 5 sudden stops and not less than 10 full turns to the right and 10 full turns to the left at the maximum towing speed of 5 mph. The jack shall be maintained under towing control at all times. At the completion of this test, the jack shall be inspected for excessive wear, distortion, or loosening of any assembly or component part.

4.6.8 Serviceability. The jack shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given to maintenance with a minimum number of tools, maintenance with general-purpose tools and equipment, servicing and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of contamination, residue, dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.6.9 Final inspection. Upon completion of all tests, the jack be disassembled to the maximum practical extent for a thorough examination of all parts to disclose failure, distortion, excessive wear, and other unsatisfactory conditions. Measurements shall be made of all parts and components to determine wear and distortion. Wear, failure, or distortion that requires rework or replacement of parts affected shall be cause for considering the affected part as having failed to complete the tests satisfactorily. Jacks not conforming to this requirement shall be rejected.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A, C or standard practice for commercial packaging (see 6.2.1.d).

5.1.1 Cleaning. Cleaning shall be in accordance with process C-1 of MIL-P-116.

5.1.2 Level A. Unless otherwise specified in the contract or order, each jack shall be preserved and packaged in accordance with MIL-P-116, method I, with applicable preservative compound or lubricant. Each jack shall be in a disassembled condition with jack fully retracted. The tow bar shall be secured in an upright position to the jack leg. Attaching parts accompanying the disassembled jack shall be preserved, bagged, or otherwise secured, appropriately identified, and attached to, or adjacent to, the intended fitting.

5.1.3 Level C. The Level C preservation for the jack shall conform to the MIL-STD-794 requirements for this level.

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* 5.1.4 Commercial. Commercial packaging shall be in accordance with ASTM D 3951.

* 5.2 Packing. Packing shall be Level A, B, C or Commercial (see 6.2.1.d).

5.2.1 Level A. The jack, preserved and packaged as specified in 5.1.2, shall be packed in an export type crate conforming to MIL-C-104, Type I, class 2, style A. Strapping and closure shall be in accordance with the appendix to the crate specification.

* 5.2.2 Level B. The jack, preserved and packaged as specified in 5.1.2, shall be packed in a domestic open crate conforming to MIL-C- 52950, Type V, Style A.

* 5.2.3 Level C. The Level C packing for jacks shall conform to the MIL-STD-794 requirements for this level.

* 5.2.4 Commercial. The jack shall be packed in accordance with the requirements of ASTM D 3951.

* 5.3 Physical protection. Interior blocking, bracing, and cushioning, for Level A and B packing shall be in accordance with the requirements of MIL-STD-1186. Level C and commercial interior blocking shall be the minimum required to protect the item, and may be the manufacturer's commercial practive provided it meets this requirement.

5.4 Marking. Unless otherwise specified in the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Jack, Hydraulic, Tripod MMU-107/E.

6. NOTES

6.1 Intended use. The MMU-107/E jack is intended for use on C-141A aircraft.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. In the event it becomes necessary to change the drawings in order to achieve the performance requirements of the specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes shall be processed in accordance with the provisions of the changes clause of this contract (see 3.2)

c. Location and conditions for first article testing (see 4.4)

d. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

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6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

6.4 Definition.

6.4.1 Cycle. One cycle consists of raising and lowering the hydraulic ram, under rated capacity load, from fully retracted position to fully extended position and returned to the fully retracted position.

* 6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-J-38475B(USAF)	Jack, Hydraulic, Tripod MMU-107/E; 40-Ton Capacity

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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